M3: Complete Implementation Retail Customer Segmentation and Pattern Discovery

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***Abstract*—This M3 milestone presents the complete implementation of a retail data mining pipeline using the UCI Online Retail dataset. The analysis integrates customer segmentation using RFM features, clustering with and without PCA, interpretable modeling using decision trees and Naive Bayes discovery for Discovery + PCA Visualization, and anomaly detection. The goal is to identify meaningful customer segments, understand purchasing behavior, and detect unusual customers. The results reveal clear behavioral patterns and provide actionable business insights.**

I. Implementation Status

The data mining pipeline is fully implemented and functional. The following components are completed:

- Data cleaning: removal of missing values, invalid transactions, and return invoices
- Feature engineering: construction of Recency, Frequency, and Monetary (RFM) features
- Customer segmentation using K-Means clustering
- Dimensionality reduction using PCA for clustering and visualization
- Model interpretation using decision trees
- Probabilistic analysis using Naive Bayes+ PCA Visualization
- Anomaly detection using Isolation Forest and LOF

All stages of the pipeline produce meaningful outputs and support the discovery objectives.

II. Analytical Choices

A. RFM-based Segmentation

RFM features were used to represent customer behavior. This approach captures recency of activity, purchase frequency, and total spending, which are essential for customer segmentation.

B. Clustering with and without PCA

K-Means clustering was applied on both original RFM features and PCA-transformed features. PCA was used as a preprocessing step to reduce feature correlation and noise, improving cluster separability.

C. Visualization Strategy

Two levels of visualization were used:

- Pairwise RFM scatter plots (e.g., Frequency vs Monetary) for local interpretation
- PCA-based plots for global structure and cluster separation

D. Decision Tree for Interpretation

Decision trees were used to explain cluster formation. The goal was not prediction accuracy but understanding which features define each customer segment.

E. Naive Bayes for Discovery

Naive Bayes was applied to analyze conditional probabilities of features given cluster labels. This provides insight into which feature ranges are most strongly associated with each segment.

F. Anomaly Detection

Isolation Forest and LOF were used to detect unusual customers, including high-value outliers and potential wholesale buyers.

III. Results

A. Customer Segmentation

K-Means clustering revealed four distinct customer segments:

- Cluster 2: VIP / Elite customers (high frequency, high spending, recent activity)
- Cluster 3: Loyal high-value customers
- Cluster 0: Regular mid-value customers
- Cluster 1: Low-value or inactive customers

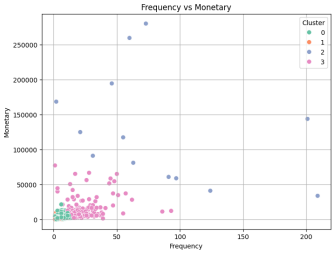

Fig. 1. Frequency VS Monetary

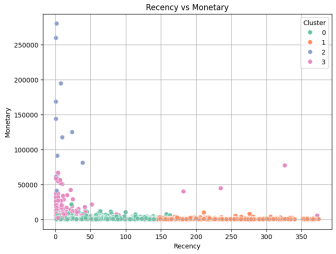

Fig. 2. Recency VS Monetary

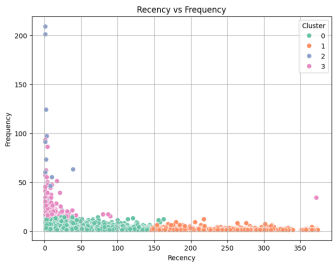

Fig. 3. Recency VS Frequency

Cluster	CustomerID	Recency	Frequency	Monetary	anomaly_score	lof_anomaly	combined_anomaly	Cluster_PCA
0	15289.128356254094	43.7026850032744	3.88271119842829 1	359.049283889802	1.0	0.996725605762933 9	0.996725605762933 9	0.002619515398652 9143
1	15347.791940016744	248.0759137769447	1.552014995313964 4	480.617479850046 9	0.998125585754451 8	0.951265229615745 1	0.951265229615745 1	0.998125585754451 8
2	15435.0	7.384615384615385	82.53846153846153	127338.31384615385	-1.0	-0.076923076923076 93	-1.0	2.076923076923077
3	15212.857843137255	15.5	22.333333333333332	12709.090490196078	0.705882352941176 5	0.941176470588235 3	0.705882352941176 5	2.470588235294117 8

Fig. 4. Cluster Summary

Analysis: 1.Cluster 2 Recency: 7 days Frequency: 82 purchases Monetary: £127,338 Characteristics: Purchased very recently Extremely high purchase frequency Exceptionally high total spending So, it shows that High-Value Loyal Customers Recommended Marketing Strategies: Exclusive VIP discounts Premium product recommendations Loyalty or membership programs Early access to new product launches .

2.Cluster 3 Recency: 15 days Frequency: 22 purchases Monetary: £12,709 Characteristics: Purchase relatively frequently High spending level Still recently active It shows that Strong Loyal Customers Although not as extreme as Cluster 2, they remain a core customer segment and contribute significantly to revenue.

3.Cluster 0 Recency: 43 days Frequency: 3.6 purchases Monetary: £1,359 Characteristics: Purchase occasionally Moderate spending Have not purchased very recently It shows that Occasional / Moderate Customers They may respond well to targeted promotions or personalized offers to increase purchase frequency.

4.Cluster 1 Recency: 248 days Frequency: 1.5 purchases Monetary: £480 Characteristics: Long time since last purchase Very low purchase frequency Lowest spending among all clusters It shows that Churned or At-Risk Customers They may have already disengaged from the business. **In addition:**Cluster 2 has a Monetary value of £127,338, which is significantly higher than all other segments.So we consider those customers are Wholesale buyers,Corporate clients Or potential outliers This results highlights the presence of a small group of customers who contribute disproportionately to total revenue.

B. Effect of PCA

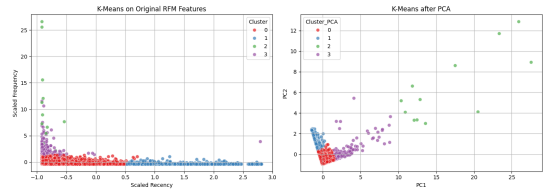

Fig. 5. PCA Comparison

PCA improves cluster visualization and separation. Compared to clustering on raw features, PCA-based clustering produces more compact and distinguishable clusters. The PCA plot provides a global view of customer structure, while RFM pair plots provide detailed interpretation.

C. Decision Tree Insights

Decision tree rules show that Frequency and Monetary are the most important features in distinguishing high-value customers, while Recency is critical for identifying inactive customers.

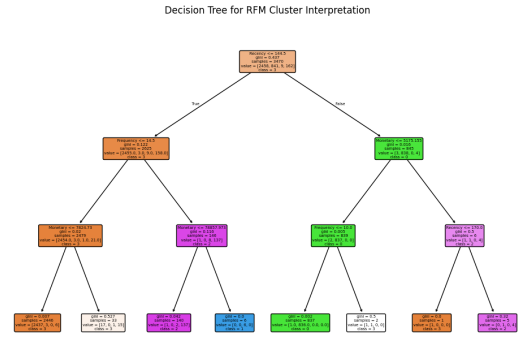

Fig. 6. Decision Tree

Rule 1 IF Recency ¡= 144.50 AND Frequency ¡= 14.50 AND Monetary ¡= 7824.73 THEN Cluster distribution: [[0.996, 0.001, 0.0, 0.002]] Interpretation: Customers who purchased recently (¡145 days ago), not very frequently (15 times), and moderate spending (£7,824) are overwhelmingly in Cluster 0 (Regular Mid-Value Customers). Business insight: These are typical occasional customers; can target with promotions to increase purchase frequency.

Rule 2 IF Recency ¡= 144.50 AND Frequency ¡= 14.50 AND Monetary ¿ 7824.73 THEN Cluster distribution: [[0.515, 0, 0.03, 0.454]] Interpretation: Recent but low-frequency customers with high spending are split between Cluster 0 and Cluster 3 (Loyal High-Value Customers). Business insight: Likely bulk or wholesale buyers — may need special engagement or VIP offers.

Rule 3 IF Recency ¡= 144.50 AND Frequency ¿ 14.50 AND Monetary ¡= 78857.97 THEN Cluster distribution: [[0.007,

0, 0.014, 0.979]] Interpretation: Frequent purchasers with moderate to high spending belong almost entirely to Cluster 3 (Loyal High-Value Customers). Business insight: Core loyal customers; focus on retention and personalized recommendations.

Rule 4 IF Recency ¡= 144.50 AND Frequency ¿ 14.50 AND Monetary ¿ 78857.97 THEN Cluster distribution: [[0, 0, 1, 0]] Interpretation: Customers with very frequent purchases and extremely high spending are Cluster 2 (VIP / Elite Customers). Business insight: These are top-tier VIPs; offer exclusive discounts, early product access, and high-end services.

Rule 5 IF Recency ¿ 144.50 AND Monetary ¡= 5175.16 AND Frequency ¡= 10.00 THEN Cluster distribution: [[0.001, 0.999, 0, 0]] **Interpretation:** Customers who haven't purchased in a long time (¿145 days), with low spending and low frequency, are almost entirely Cluster 1 (Lost / Low-Value Customers). Business insight: At-risk customers; may need re-engagement campaigns or win-back promotions.

Rule 6 IF Recency ¿ 144.50 AND Monetary ¡= 5175.16 AND Frequency ¿ 10.00 THEN Cluster distribution: [[0.5, 0.5, 0, 0]] Interpretation: Long time since last purchase but relatively frequent historical buyers — split between Cluster 0 and 1. Business insight: Could indicate churned regulars; track reactivation opportunities.

Rule 7 IF Recency ¿ 144.50 AND Monetary ¿ 5175.16 AND Recency ¡= 170.00 THEN Cluster distribution: [[1, 0, 0, 0]] Interpretation: Customers who purchased somewhat recently, with high spending, are classified in Cluster 0 (Mid-Value). Business insight: Likely intermediate buyers or occasional bulk customers.

Rule 8 IF Recency ¿ 144.50 AND Monetary ¿ 5175.16 AND Recency ¿ 170.00 THEN Cluster distribution: [[0, 0.2, 0, 0.8]] Interpretation: Customers who haven't purchased for a long time, but had high spending, are mainly Cluster 3 (Loyal High-Value) with some Cluster 1. Business insight: Valuable but dormant customers; prime candidates for VIP reactivation campaigns.

D. Naive Bayes for Discovery + PCA Visualization

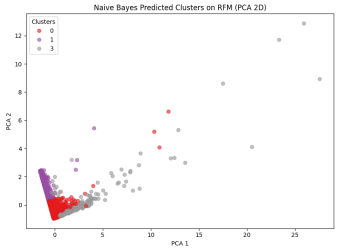

Fig. 7. Naive Bayes Predicted Clusters on RFM

Naive Bayes Results:

1. Class Priors

The class priors show how common each cluster is in the dataset. Cluster 0 has about 70

2. Conditional Probabilities

The conditional probabilities help us understand how RFM features relate to each cluster.

Recency: Cluster 2 customers usually purchased very recently. Cluster 0 customers have moderate recency. Cluster 1 customers have not purchased for a long time.

Frequency: Cluster 2 customers buy very frequently. Cluster 0 customers buy occasionally. Cluster 1 customers rarely make purchases.

Monetary: Cluster 2 customers spend the most (VIP level). Cluster 0 customers have moderate spending. Cluster 1 customers spend very little. This confirms that Cluster 2 are VIP/Elite customers, Cluster 3 are loyal high-value customers, Cluster 1 are low-value or inactive, and Cluster 0 are regular customers.

3. Classification Report

The overall accuracy is about 87

4. Discovery Insights

Cluster 2 customers are VIPs: they buy recently, frequently, and spend the most. Cluster 1 customers are at risk or inactive and may need re-engagement campaigns. Cluster 0 customers are regular buyers and could be targeted with upselling or cross-selling. The probability patterns can be used to design simple marketing rules or recommendations.

- VIP customers are strongly associated with high Frequency and Monetary bins
- Low-value customers are associated with high Recency (long inactivity)
- Moderate customers fall into middle feature ranges

PCA visualization of Naive Bayes predictions shows general separation of clusters, although rare clusters (VIP customers) are underrepresented due to class imbalance.

E. Anomaly Detection

The goal is to identify unusual customers, such as extreme VIPs, potential wholesalers, or anomalies. We'll use Isolation Forest (robust for high-value outliers) and Local Outlier Factor (LOF) for comparison.

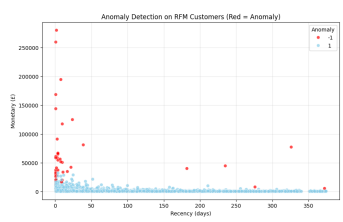

Fig. 8. Anomaly Detection on RFM

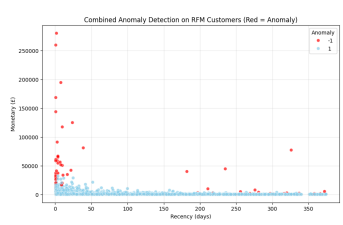

Fig. 9. Combined Anomaly Detection on RFM

As the dataset, it seems that the really extreme values in Monetary and Frequency are having the biggest impact on finding anomalies. Both Isolation Forest and LOF end up flagging the same customers with either very high spending or unusual buying patterns, so their results overlap a lot. A few observations from the top anomalies:

CustomerID 14646 and 18102 – very low Recency (2 days or 1 day), very high Frequency or Monetary → likely VIP or wholesale buyers. CustomerID 14911 – extreme Frequency of 201 → probably bulk purchases, maybe company accounts. CustomerID 12346 – extreme Recency of 326 days but low Monetary → could be long-dormant customer or one-time bulk buyer. Main takeaway: The anomaly detection is working as expected. It points out both super high-value customers and those with unusual buying behavior.

- Very high-spending customers (potential wholesale buyers)
- Customers with unusually high purchase frequency
- Rare behavioral patterns

These customers contribute disproportionately to revenue and require targeted strategies.

IV. Discovery Question Progress

The analysis addresses the discovery questions from M1:

1. Product Co-purchase Behavior: Association rules (from M2) identify frequent product combinations, supporting recommendation systems and cross-selling strategies.

2. Customer Segmentation: RFM-based clustering reveals clear and interpretable customer segments.

3. Anomaly Detection: Anomaly detection methods successfully identify unusual customers, including high-value and rare behavior patterns.

Overall, the discovery questions are largely answered with strong supporting evidence.

V. Plan for Milestone 4

The next milestone will focus on improving presentation and analysis quality:

- Enhance visualization clarity and figure integration
- Provide deeper interpretation of customer segments
- Evaluate model limitations and robustness
- Compare additional clustering methods (e.g., DBSCAN)
- Improve narrative flow

The implementation is complete; remaining work focuses on refinement and presentation.